Mandatory deliverable

LAB | Host a Static Website on S3 Lab

Introduction

In this hands-on lab, you will create an Amazon S3 bucket, upload multiple versions of a file, and configure the bucket to host a static website. Additionally, you will apply bucket policies to enable public access.

Objectives

By the end of this lab, you will be able to:

1. Create and configure an S3 bucket.
2. Enable versioning and upload multiple file versions.
3. Configure static website hosting for the S3 bucket.
4. Apply bucket policies to allow public access.
5. Verify accessibility of the hosted website.
6. Manage and clean up resources after completion.

Lab Instructions

1. Create an S3 Bucket

- Navigate to the **AWS S3 Console**.
- Click **Create Bucket**.
- Enter a **unique bucket name** (e.g., `my-static-site-bucket`).
- Uncheck **"Block all public access"** to allow public visibility.
- Enable **Bucket Versioning**.
- Click **Create Bucket**.

2. Upload File Versions

- On your local machine, create an HTML file named `index.html` with the following content:

```
<h1>Welcome to My Static Website!</h1>
```

Ask AI Tutor

Copy Hide explanation

This line creates the main visible heading for a web page. In HTML, `<h1>` means "top-level heading," so the browser will display this text larger and bolder than normal text to show it is the primary title of the page.

The important idea is that static website content in S3 is just ordinary files like this HTML file. When someone opens the site, the browser reads the file and renders the heading exactly as written.

One useful mental model: HTML tags describe structure, not just appearance. Here, the tag tells the browser, "this is the page's main title," which is why it stands out automatically.

WAS THIS HELPFUL?

AI TUTOR ACTIONS

Ask a follow-up Explain more simply Show another example

- Upload the `index.html` file to your **S3 bucket**.
- Navigate to the **"Versions"** tab to confirm the initial upload.
- Modify `index.html` locally to:

```
<h1>Updated Version of My Static Website!</h1>
```

Copy Hide explanation

This HTML snippet is the content of a webpage heading. The `<h1>` tag means "top-level heading," so the browser will display this text as the main title of the page, usually in a large, bold style.

The important idea here is that static website hosting on S3 serves files exactly as they are stored. If this file is uploaded as `index.html`, then visiting the website endpoint will show this heading in the browser. Changing the text inside the tag changes what visitors see, which is why uploading a new version updates the site content.

One useful mental model: S3 is not "running" this code like an app server would; it is simply delivering an HTML file. So the page only changes when the file content changes and the updated file is uploaded.

WAS THIS HELPFUL?

AI TUTOR ACTIONS

Ask a follow-up Explain more simply Show another example

- Upload the new version of `index.html` to the **S3 bucket**.

3. Configure Static Website Hosting

- Open the **Properties** tab of your **S3 bucket**.
- Locate **"Static website hosting"** and select **Enable**.
- Set the **index document** as `index.html`.
- Click **Save**.
- Copy the **S3 static website endpoint URL** for later testing.

4. Set Bucket Policy for Public Access

- Navigate to the **Permissions** tab.
- Click **Edit Bucket Policy** and paste the following policy:

Copy | Hide explanation

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-static-site-bucket/*"
    }
  ]
}
```

This JSON is an S3 bucket policy, and its job is to make objects in the bucket publicly readable. In other words, it tells AWS: "allow anyone on the internet to download files from this bucket," which is what a static website needs so browsers can fetch `index.html` and other assets.

`Version` identifies the policy language format AWS should interpret. `Statement` is the list of permission rules, and here there is only one rule. `Sid` is just a label for that rule, useful for readability and troubleshooting.

The important part is the permission logic: `Effect: "Allow"` grants access, `Principal: "*"` means any user or service, `Action: "s3:GetObject"` allows reading objects, and `Resource: "arn:aws:s3:::my-static-site-bucket/*"` limits that permission to every object inside that specific bucket. The `/*` matters because it targets the files in the bucket, not the bucket itself.

A key mental model: this policy does not make the bucket "fully public" in every way; it only allows public read access to objects. If the bucket name doesn't match your actual bucket, or if public access is still blocked elsewhere in S3 settings, the website can still fail with a 403 error.

WAS THIS HELPFUL?

AI TUTOR ACTIONS

Ask a follow-up | Explain more simply | Show another example

- **Replace** `my-static-site-bucket` with your actual bucket name.
- Click **Save Policy**.

5. Verify Website Access

- Open a browser and enter the **S3 static website URL**.
- Ensure that the **HTML content loads correctly**.
- If a **403 Forbidden** error appears, check your **Bucket Policy and Object ACLs**.

6. Upload and Test File Versioning

- Modify the `index.html` file again with new content.
- Upload the updated file to the **S3 bucket**.
- Refresh the **S3 static website URL** to confirm changes.

7. Cleanup Resources

- **Empty the S3 bucket** by deleting all objects.
- **Disable static website hosting**.
- **Delete the S3 bucket** to prevent unnecessary costs.

Completion Criteria

- S3 bucket created and configured with versioning enabled.
- Multiple versions of `index.html` uploaded successfully.
- Static website hosting enabled and accessible.
- Bucket policy applied to allow public read access.
- Final verification of website functionality.
- Cleanup completed.

Submission Guidelines

- Take **screenshots** of each major step.
- Save them into a **document (Google Docs, PDF, etc.)**.
- Upload to **Google Drive** and share the link.
- Ensure **instructor has access** before submission.

🚀 **Congratulations! You've successfully hosted a static website on S3!** 🎉

LAB | Host a Static Website on S3

Lab

URL

Enter the URL here

Submit lab

Previous

Week 2: Day 3
LAB | Amazon DynamoDB Basics

Next